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DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING



NOTES

Subject with code: Electronic Principles and Circuits (BEC303)

Module 5

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Syllabus :

Power Amplifiers: Amplifier terms, Two load lines, Class A Operation, Class B operation, Class B push pull emitter follower, Class C Operation.

Thyristors: The four layer Diode, SCR, SCR Phase control, Bidirectional Thyristors, IGBTs, Other Thyristors.

Amplifier Terms

There are different ways to describe amplifiers. For instance, we can describe them by their class of operation, by their interstage coupling, or by their frequency range.

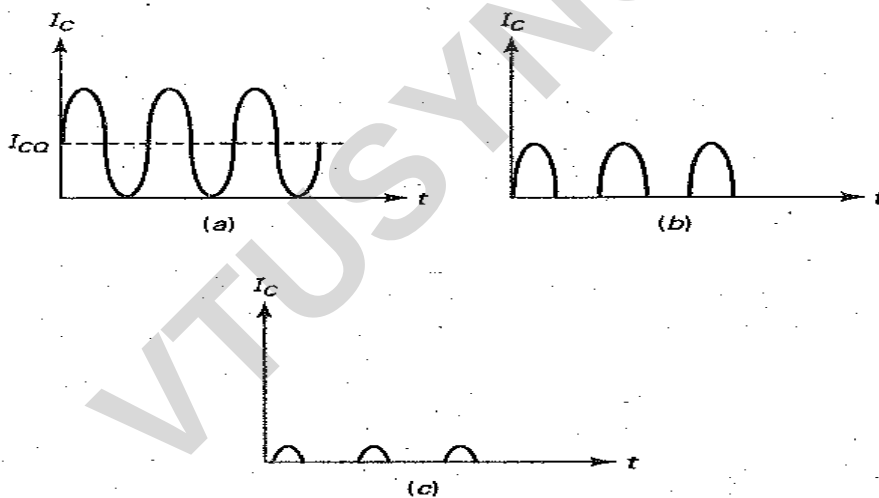
Classes of operation

Class A operation of an amplifier means that the transistor operates in the active region at all times. This implies that collector current flows for 360° of the ac cycle, as shown in Fig. 12-1a. With a class A amplifier, the designer usually tries to locate the Q point somewhere near the middle of the load line. This way, the signal can swing over the maximum possible range without saturating or cutting off the transistor, which would distort the signal.

Class B operation is different. It means that collector current flows for only half the cycle (180°), as shown in Fig. 12-1b. To have this kind of operation, a designer locates the Q point at cutoff. Then, only the positive half of ac base voltage can produce collector current. This reduces the wasted heat in power transistors.

Class C operation means that collector current flows for less than 180° of the ac cycle, as shown in Fig. 12-1c. With class C operation, only part of the positive half cycle of ac base voltage produces collector current. As a result, we get brief pulses of collector current like those of Fig. 12-1c.

Figure 12-1 Collector current: (a) class A; (b) class B; (c) class C.



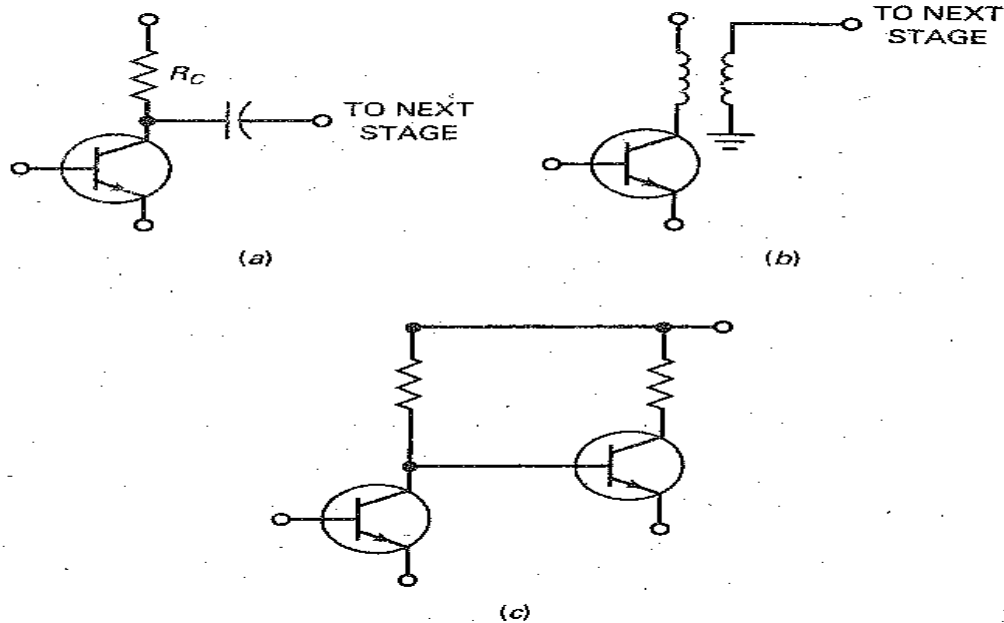
Types of Coupling

Figure 12-2a shows **capacitive coupling**. The coupling capacitor transmits the amplified ac voltage to the next stage. Figure 12-2b illustrates **transformer coupling**. Here the ac voltage is coupled through a transformer to the next stage. Capacitive coupling and transformer coupling are both examples of ac coupling, which blocks the dc voltage.

Direct coupling is different. In Fig. 12-2c, there is a direct connection between the collector of the first transistor and the base of the second transistor.

Because of this, both the dc and the ac voltages are coupled. Since there is no lower frequency limit, a direct-coupled amplifier is sometimes called a *dc amplifier*.

Figure 12-2 Types of coupling: (a) capacitive; (b) transformer; (c) direct.



Range of frequency

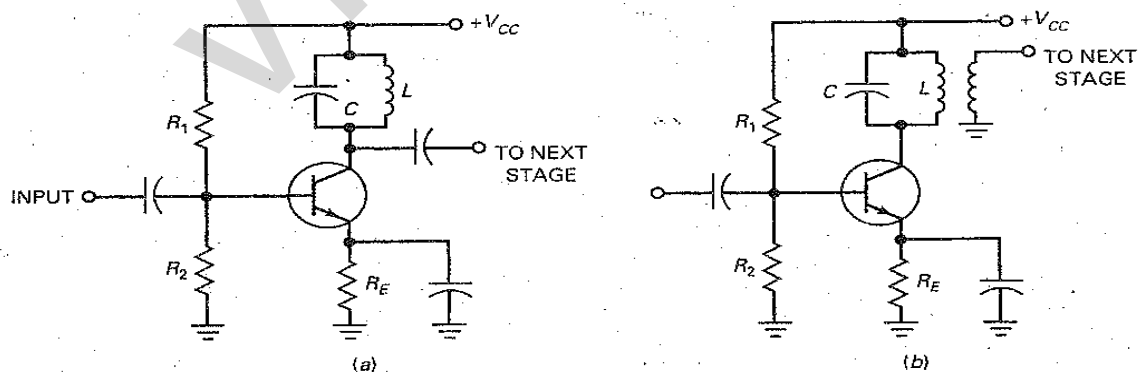
Another way to describe amplifiers is by stating their frequency range. For instance, an **audio amplifier** refers to an amplifier that operates in the range of 20 Hz to 20 kHz. On the other hand, a **radio-frequency (RF) amplifier** is one that amplifies frequencies above 20 kHz, usually much higher.

Amplifiers are also classified as **narrowband** or **wideband**. A narrowband amplifier works over a small frequency range like 450 to 460 kHz. A wideband amplifier operates over a large frequency range like 0 to 1 MHz.

Figure 12-3a is an example of a tuned RF amplifier. The LC tank is resonant at some frequency. If the tank has a high Q , the bandwidth is narrow. The output is capacitively coupled to the next stage.

Figure 12-3b is another example of a tuned RF amplifier. This time, the narrowband output signal is transformer-coupled to the next stage.

Figure 12-3 Tuned RF amplifiers: (a) capacitive coupling; (b) transformer coupling.



Signal Levels

We have already described *small-signal operation*, in which the peak-to-peak swing in collector current is less than 10 percent of quiescent collector current. In **large-signal operation**, a peak-to-peak signal uses all or most of the load line. In a stereo system, the small signal from a radio tuner, tape player, or compact disc

player is used as the input to a **preamp**, an amplifier that produces a larger output suitable for driving tone and volume controls. The signal is then used as the input to a **power amplifier**, which produces output power ranging from a few hundred milliwatts up to hundreds of watts.

Two load lines

DC load line

Figure 12-4a is a voltage-divider-based (VDB) amplifier. One way to move the Q point is by varying the value of R_2 . For very large values of R_2 , the transistor goes into saturation and its current is given by:

$$I_{C(\text{sat})} = \frac{V_{CC}}{R_C + R_E} \quad (12-1)$$

Very small values of R_2 will drive the transistor into cutoff, and its voltage is given by:

$$V_{CE(\text{cutoff})} = V_{CC} \quad (12-2)$$

Figure 12-4b shows the dc load line with the Q point.

AC load line

Figure 12-4c is the ac equivalent circuit for the VDB amplifier. With the emitter at ac ground, R_E has no effect on the ac operation. Furthermore, the ac collector resistance is less than the dc collector resistance. Therefore, when an ac signal comes in, the instantaneous operating point moves along the **ac load line** of Fig. 12-4d. In other words, the peak-to-peak sinusoidal current and voltage are determined by the ac load line.

Here's how to determine the ends of the ac load line. Writing a collector voltage loop gives us:

$$v_{ce} + i_c r_c = 0$$

or

$$i_c = -\frac{v_{ce}}{r_c} \quad (12-3)$$

The ac collector current is given by:

$$i_c = \Delta I_C = I_C - I_{CQ}$$

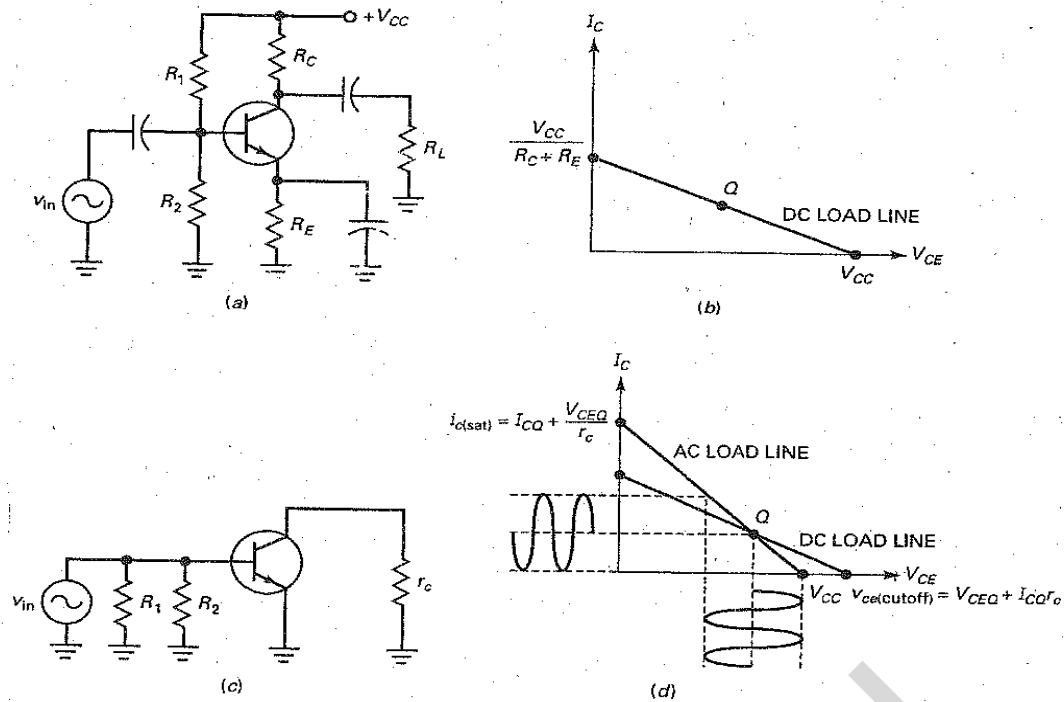
and the ac collector voltage is:

$$v_{ce} = \Delta V_{CE} = V_{CE} - V_{CEQ}$$

When substituting these expressions into Eq. (12-3) and rearranging, we arrive at:

$$I_C = I_{CQ} + \frac{V_{CEQ}}{r_c} - \frac{V_{CE}}{r_c} \quad (12-4)$$

Figure 12-4 (a) VDB amplifier; (b) dc load line; (c) ac equivalent circuit; (d) ac load line.



This is the equation of the ac load line. When the transistor goes into saturation, V_{CE} is zero, and Eq. (12-4) gives us:

$$i_{c(sat)} = I_{CQ} + \frac{V_{CEQ}}{r_c} \quad (12-5)$$

where $i_{c(sat)}$ = ac saturation current

I_{CQ} = dc collector current

V_{CEQ} = dc collector-emitter voltage

r_c = ac resistance seen by the collector

When the transistor goes into cutoff, I_c equals zero. Since

$$v_{ce(cutoff)} = V_{CEQ} + \Delta V_{CE}$$

and

$$\Delta V_{CE} = (\Delta I_C)(r_c)$$

we can substitute to get:

$$\Delta V_{CE} = (I_{CQ} - OA)(r_c)$$

resulting in:

$$v_{ce(cutoff)} = V_{CEQ} + I_{CQ}r_c \quad (12-6)$$

Because the ac load line has a higher slope than the dc load line, the maximum peak-to-peak (MPP) output is always less than the supply voltage. As a formula:

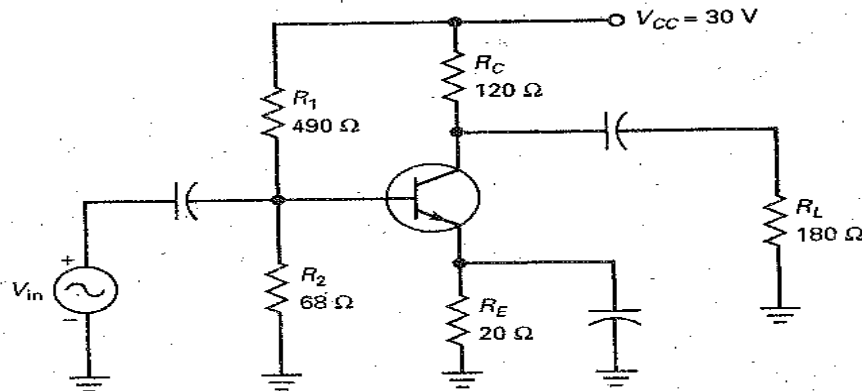
$$\text{MPP} < V_{CC} \quad (12-7)$$

Example 12-1

III Multisim

What are the values of I_{CQ} , V_{CEQ} and r_c in Fig. 12-7?

Figure 12-7 Example.



SOLUTION

$$V_B = \frac{68 \, \Omega}{68 \, \Omega + 490 \, \Omega} (30 \, \text{V}) = 3.7 \, \text{V}$$

$$V_E = V_B - 0.7 \, \text{V} = 3.7 \, \text{V} - 0.7 \, \text{V} = 3 \, \text{V}$$

$$I_E = \frac{V_E}{R_E} = \frac{3 \, \text{V}}{20 \, \Omega} = 150 \, \text{mA}$$

$$I_{CQ} \cong I_E = 150 \, \text{mA}$$

$$V_{CEQ} = V_C - V_E = 12 \, \text{V} - 3 \, \text{V} = 9 \, \text{V}$$

$$r_c = R_C \parallel R_L = 120 \, \Omega \parallel 180 \, \Omega = 72 \, \Omega$$

Example 12-2

Determine the ac load line saturation and cutoff points in Fig. 12-7. Also, find the maximum peak-to-peak output voltage.

SOLUTION From Example 12-1, the transistor's Q point is:

$$I_{CQ} = 150 \, \text{mA} \quad \text{and} \quad V_{CEQ} = 9 \, \text{V}$$

To find the ac saturation and cutoff points, first determine the ac collector resistance, r_c :

$$r_c = R_C \parallel R_L = 120 \, \Omega \parallel 180 \, \Omega = 72 \, \Omega$$

Next find the ac load line end points:

$$i_{c(\text{sat})} = I_{CQ} + \frac{V_{CEQ}}{r_c} = 150 \, \text{mA} + \frac{9 \, \text{V}}{72 \, \Omega} = 275 \, \text{mA}$$

$$v_{ce(\text{cutoff})} = V_{CEQ} + I_{CQ}r_c = 9 \, \text{V} + (150 \, \text{mA})(72 \, \Omega) = 19.8 \, \text{V}$$

Now determine the MPP value. With a supply voltage of 30 V:

$$\text{MPP} < 30 \, \text{V}$$

MP will be the smaller value of:

$$I_{CQ}r_c = (150 \, \text{mA})(72 \, \Omega) = 10.8 \, \text{V}$$

or

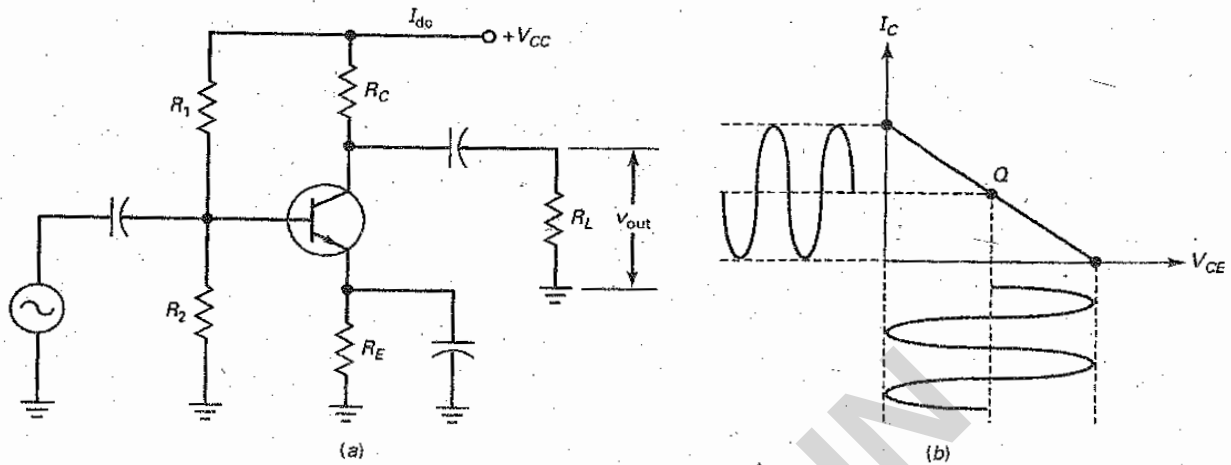
$$V_{CEQ} = 9 \, \text{V}$$

$$\text{Therefore, MPP} = 2(9 \, \text{V}) = 18 \, \text{V}$$

Class A Operation

The VDB amplifier of Fig. 12-8a is a class A amplifier as long as the output signal is not clipped. With this kind of amplifier, collector current flows throughout the cycle. Stated another way, no clipping of the output signal occurs at any time during the cycle.

Figure 12-8 Class A amplifier.



Power gain of Class A amplifier

Besides voltage gain, any amplifier has a **power gain**, defined as:

$$A_p = \frac{P_{out}}{P_{in}} \quad (12-12)$$

In words, the power gain equals the ac output power divided by the ac input power.

Output power of class A amplifier

If we measure the output voltage of Fig. 12-8a in rms volts, the output power is given by

$$P_{out} = \frac{v_{rms}^2}{R_L} \quad (12-13)$$

Usually, we measure the output voltage in peak-to-peak volts with an oscilloscope. In this case, a more convenient equation to use for output power is:

$$P_{out} = \frac{v_{out}^2}{8R_L} \quad (12-14)$$

The factor of 8 in the denominator occurs because $v_{pp} = 2\sqrt{2} v_{rms}$. When you square $2\sqrt{2}$, you get 8.

The maximum output power occurs when the amplifier is producing the maximum peak-to-peak output voltage, as shown in Fig. 12-8b. In this case, v_{pp} equals the maximum peak-to-peak output voltage and the maximum output power is:

$$P_{out(max)} = \frac{MPP^2}{8R_L} \quad (12-15)$$

Transistor power dissipation

When no signal drives the amplifier of Fig. 12-8a, the quiescent power dissipation is:

$$P_{DQ} = V_{CEQ}I_{CQ} \quad (12-16)$$

This makes sense. It says that the quiescent power dissipation equals the dc voltage times the dc current.

Drain current

As shown in Fig. 12-8a, the dc voltage source has to supply a dc current I_{dc} to the amplifier. This dc current has two components: the biasing current through the voltage divider and the collector current through the transistor. The dc current is called the **current drain** of the stage. If you have a multistage amplifier, you have to add the individual current drains to get the total current drain.

Efficiency of Class A amplifier

The dc power supplied to an amplifier by the dc source is:

$$P_{dc} = V_{CC}I_{dc} \quad (12-17)$$

To compare the design of power amplifiers, we can use the **efficiency**, defined by:

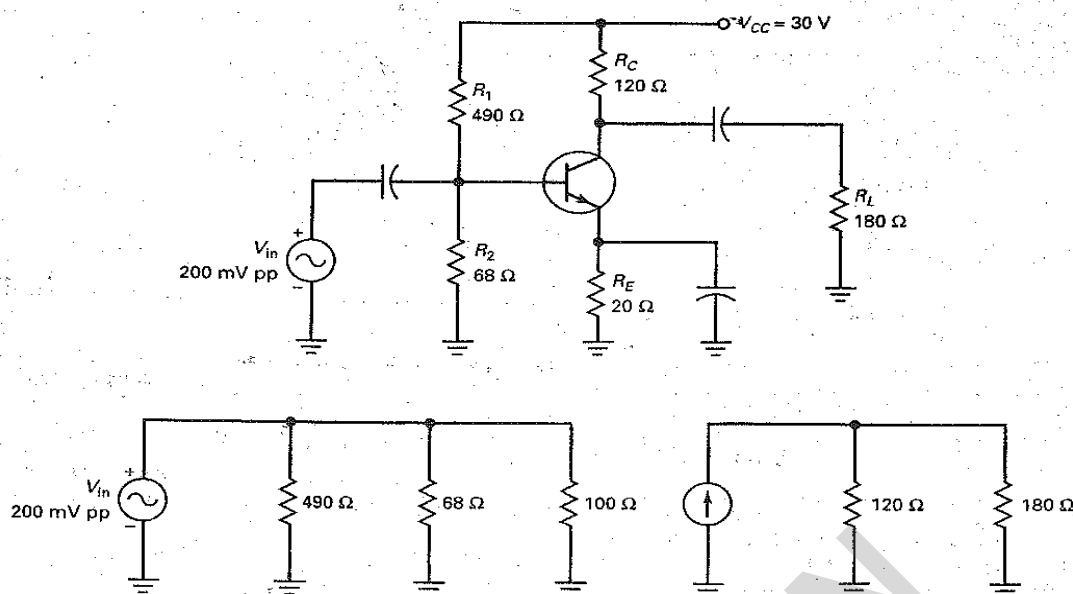
$$\eta = \frac{P_{out}}{P_{dc}} \times 100\% \quad (12-18)$$

This equation says that the efficiency equals the ac output power divided by the dc input power.

Example 12-3

If the peak-to-peak output voltage is 18 V and the input impedance of the base is 100 Ω , what is the power gain in Fig. 12-9a?

Figure 12-9 Example.



SOLUTION As shown in Fig. 12-9b:

$$Z_{in(stage)} = 490 \Omega \parallel 68 \Omega \parallel 100 \Omega = 37.4 \Omega$$

The ac input power is:

$$P_{in} = \frac{(200 \text{ mV})^2}{8 (37.4)} = 133.7 \mu\text{W}$$

The ac output power is:

$$P_{out} = \frac{(18 \text{ V})^2}{8 (180 \Omega)} = 225 \text{ mW}$$

The power gain is:

$$A_p = \frac{225 \text{ mW}}{133.7 \mu\text{W}} = 1683$$

Example 12-4

What is the transistor power dissipation and efficiency of Fig. 12-9a?

SOLUTION The dc emitter current is:

$$I_E = \frac{3 \text{ V}}{20 \Omega} = 150 \text{ mA}$$

The dc collector voltage is:

$$V_C = 30 \text{ V} - (150 \text{ mA})(120 \Omega) = 12 \text{ V}$$

and the dc collector-emitter voltage is:

$$V_{CEQ} = 12 \text{ V} - 3 \text{ V} = 9 \text{ V}$$

The transistor power dissipation is:

$$P_{DQ} = V_{CEQ} I_{CQ} = (9 \text{ V})(150 \text{ mA}) = 1.35 \text{ W}$$

To find the stage efficiency:

$$I_{\text{bias}} = \frac{30 \text{ V}}{490 \Omega + 68 \Omega} = 53.8 \text{ mA}$$

$$I_{\text{dc}} = I_{\text{bias}} + I_{CQ} = 53.8 \text{ mA} + 150 \text{ mA} = 203.8 \text{ mA}$$

The dc input power to the stage is:

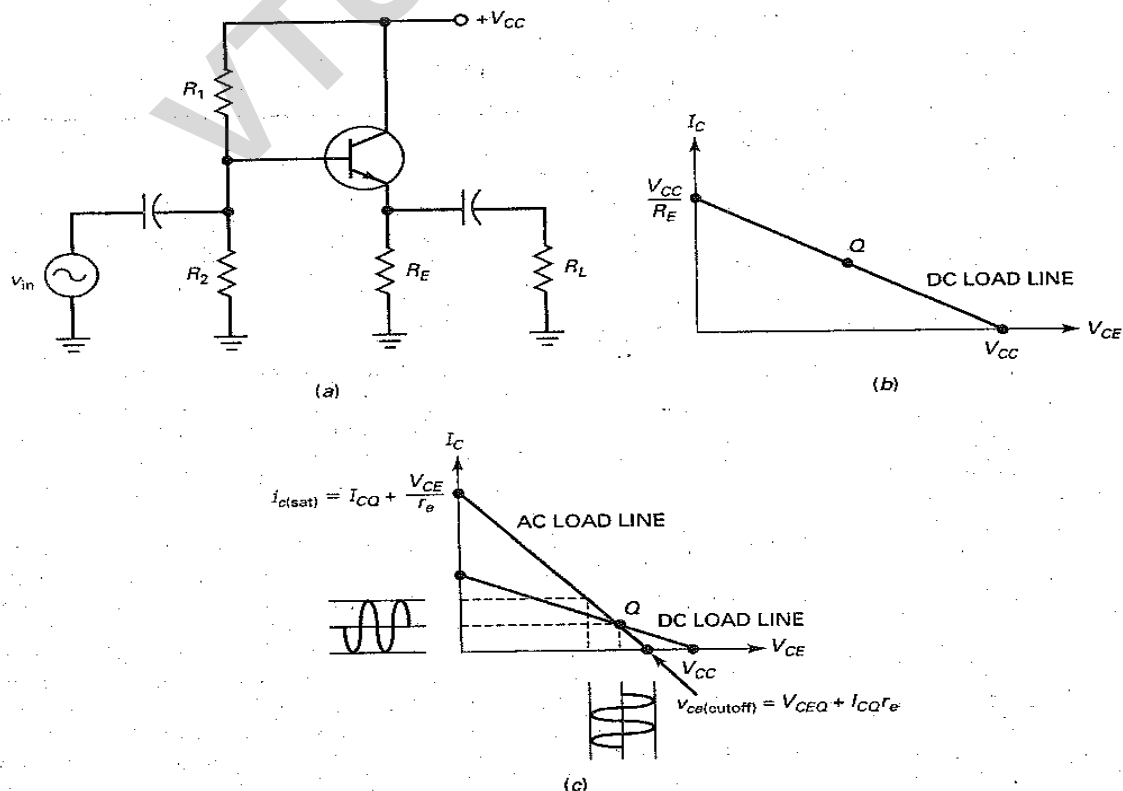
$$P_{\text{dc}} = V_{CC} I_{\text{dc}} = (30 \text{ V})(203.8 \text{ mA}) = 6.11 \text{ W}$$

Since the output power (found in Example 12-3) is 225 mW, the efficiency of the stage is:

$$\eta = \frac{225 \text{ mW}}{6.11 \text{ W}} \times 100\% = 3.68\%$$

Emitter follower power amplifier

Figure 12-11 DC and ac load lines.



In Fig. 12-11a, large values of R_2 will saturate the transistor, producing a saturation current of:

$$I_{C(sat)} = \frac{V_{CC}}{R_E} \quad (12-19)$$

Small values of R_2 will drive the transistor into cutoff, producing a cut-off voltage of:

$$V_{CE(cutoff)} = V_{CC} \quad (12-20)$$

Fig. 12-11b shows the dc load line with the Q point.

In Fig. 12-11a, the ac emitter resistance is less than the dc emitter resistance. Therefore, when an ac signal comes in, the instantaneous operating point moves along the ac load line of Fig. 12-11c. The peak-to-peak sinusoidal current and voltage are determined by the ac load line.

As shown in Fig. 12-11c, the ac load line end points are found by:

$$i_{c(sat)} = I_{CQ} + \frac{V_{CE}}{r_e} \quad (12-21)$$

and

$$V_{CE(cutoff)} = V_{CE} + I_{CQ} r_e \quad (12-22)$$

Because the ac load line has a higher slope than the dc load line, the maximum peak-to-peak output is always less than the supply voltage. As with the class A CE amplifier, $MPP < V_{CC}$.

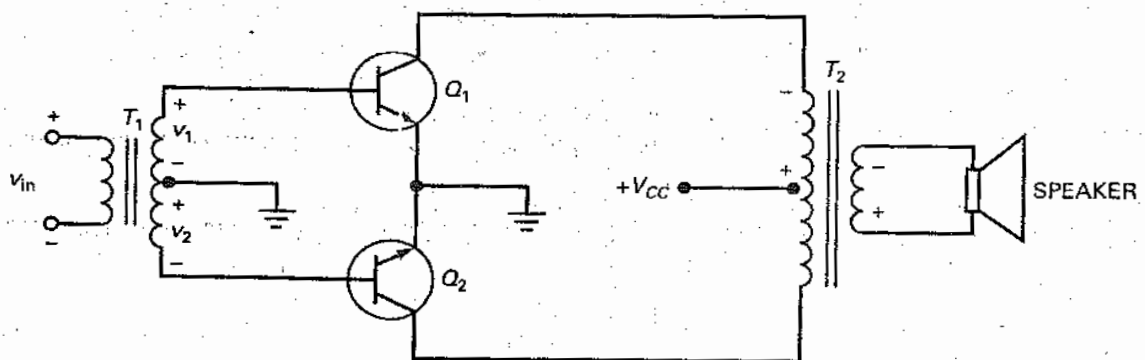
When the Q point is below the center of the ac load line, the maximum peak (MP) output is $I_{CQ} r_e$, as shown in Fig. 12-12a. On the other hand, if the Q point is above the center of the load line, the maximum peak output is V_{CEQ} , as shown in Fig. 12-12b.

Class B Operation

Class B Push Pull amplifier

Figure 12-14 shows a basic class B amplifier. When a transistor operates as class B, it clips off half a cycle. To avoid the resulting distortion, we can use two

Figure 12-14 Class B push-pull amplifier.



transistors in a push-pull arrangement like that of Fig. 12-14. **Push-pull** means that one transistor conducts for half a cycle while the other is off, and vice versa.

Here is how the circuit works: On the positive half cycle of input voltage, the secondary winding of T_1 has voltage v_1 and v_2 , as shown. Therefore, the upper transistor conducts and the lower one cuts off. The collector current through Q_1 flows through the upper half of the output primary winding. This produces an amplified and inverted voltage, which is transformer-coupled to the loudspeaker.

On the next half cycle of input voltage, the polarities reverse. Now, the lower transistor turns on and the upper transistor turns off. The lower transistor amplifies the signal, and the alternate half cycle appears across the loudspeaker.

Since each transistor amplifies one-half of the input cycle, the loudspeaker receives a complete cycle of the amplified signal.

Advantages of Class B push pull amplifier

Since there is no bias in Fig. 12-14, each transistor is at cutoff when there is no input signal, an advantage because there is no current drain when the signal is zero.

Another advantage is improved efficiency where there is an input signal. The maximum efficiency of a class B push-pull amplifier is 78.5 percent, so a class B push-pull power amplifier is more commonly used for an output stage than a class A power amplifier.

Disadvantages of Class B push pull amplifier

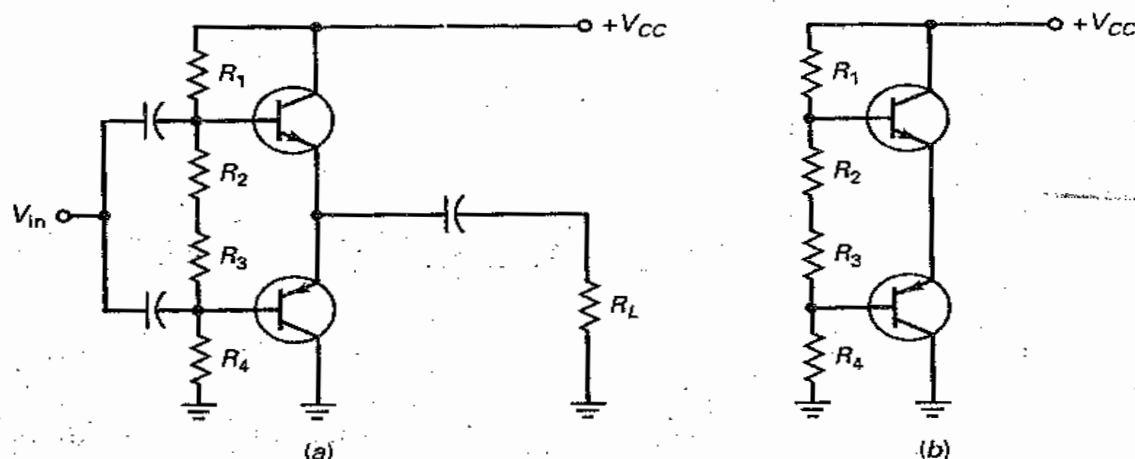
The main disadvantage of the amplifier shown in Fig. 12-14 is the use of transformers. Audio transformers are bulky and expensive. Although widely used at one time, a transformer-coupled amplifier like Fig. 12-14 is no longer popular. Newer designs have eliminated the need for transformers in most applications.

Class B push pull emitter follower

Figure 12-15a shows one way to connect a class B push-pull emitter follower. Here, we have an *npn* emitter follower and a *pnp* emitter follower connected in a push-pull arrangement.

Let's begin the analysis with the dc equivalent circuit of Fig. 12-15b. The designer selects biasing resistors to set the Q point at cutoff. This biases the

Figure 12-15 Class B push-pull emitter follower: (a) complete circuit; (b) dc equivalent circuit.



emitter diode of each transistor between 0.6 and 0.7 V, so that it is on the verge of conduction. Ideally:

$$I_{CQ} = 0$$

Because the biasing resistors are equal, each emitter diode is biased with the same value of voltage. As a result, half the supply voltage is dropped across each transistor's collector-emitter terminals. That is:

$$V_{CEQ} = \frac{V_{CC}}{2} \quad (12-23)$$

On the positive half cycle of input voltage, the upper transistor of Fig. 12-15a conducts and the lower one cuts off. The upper transistor acts like an ordinary emitter follower, so that the output voltage approximately equals the input voltage.

On the negative half cycle of input voltage, the upper transistor cuts off and the lower transistor conducts. The lower transistor acts like an ordinary emitter follower and produces a load voltage approximately equal to the input voltage. The upper transistor handles the positive half cycle of input voltage, and the lower transistor takes care of the negative half cycle. During either half cycle, the source sees a high input impedance looking into either base.

DC load line

Since there is no dc resistance in the collector or emitter circuits of Fig. 12-15b, the dc saturation current is infinite. This means that the dc load line is vertical, as shown in Fig. 12-16a. If you think that this is a dangerous situation, you are right.

AC load line

Figure 12-16 (a) DC and ac load lines; (b)

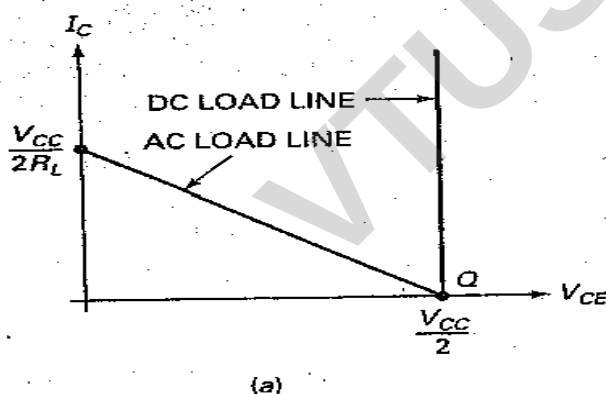


Figure 12-16a shows the ac load line. When either transistor is conducting, its operating point moves up along the ac load line. The voltage swing of the conducting transistor can go all the way from cutoff to saturation. On the alternate half cycle, the other transistor does the same thing. This means that the maximum peak-to-peak output is:

$$MPP = V_{CC} \quad (12-24)$$

Class C Operation

With class C operation, the collector current flows for less than half a cycle. A parallel resonant circuit can filter the pulses of collector current and produce a pure sine wave of output voltage. The main application for class C is with tuned RF amplifiers. The maximum efficiency of a tuned class C amplifier is 100 percent.

Figure 12-25a shows a tuned RF amplifier. The ac input voltage drives the base, and an amplified output voltage appears at the collector. The amplified and inverted signal is then capacitively coupled to the load resistance. Because of the parallel resonant circuit, the output voltage is maximum at the resonant frequency, given by:

$$f_r = \frac{1}{2\pi\sqrt{LC}} \quad (12-29)$$

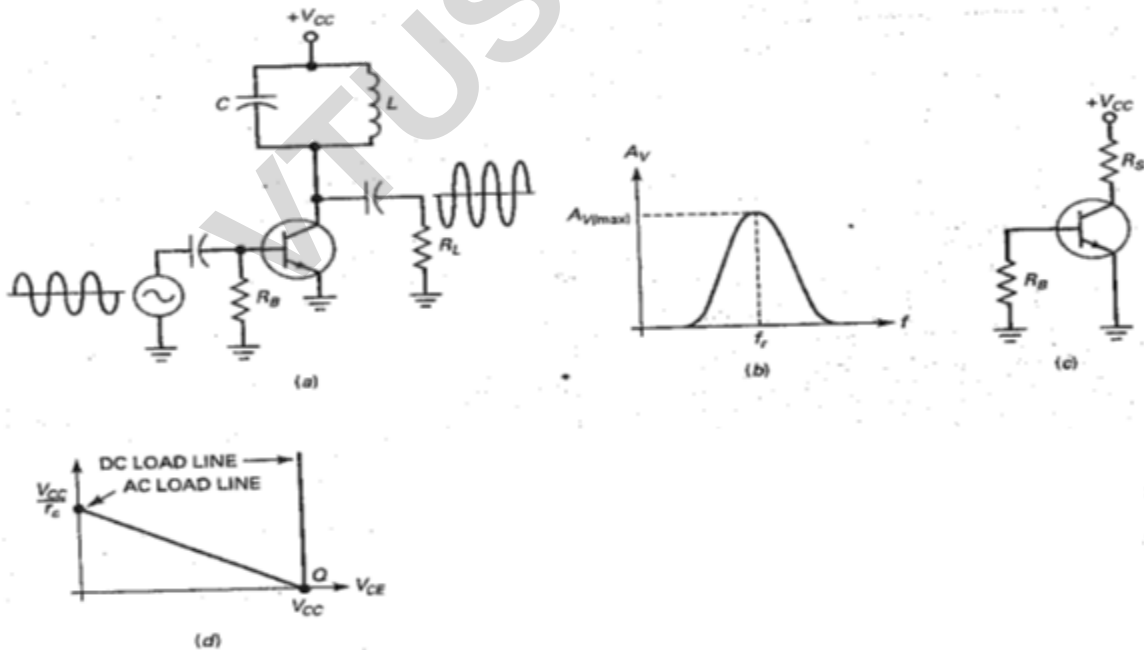
On either side of the resonant frequency f_r , the voltage gain drops off as shown in Fig. 12-25b. For this reason, a tuned class C amplifier is always intended to amplify a narrow band of frequencies. This makes it ideal for amplifying radio and television signals because each station or channel is assigned a narrow band of frequencies on both sides of a center frequency.

The class C amplifier is unbiased, as shown in the dc equivalent circuit of Fig. 12-25c. The resistance R_S in the collector circuit is the series resistance of the inductor.

Load line of class C amplifier

Figure 12-25d shows the two load lines. The dc load line is approximately vertical because the winding resistance R_S of an RF inductor is very small. The dc load line is not important because the transistor is unbiased. What is important is the ac load line. As indicated, the Q point is at the lower end of the ac load line. When an ac signal is present, the instantaneous operating point moves up the ac load line toward the saturation point. The maximum pulse of collector current is given by the saturation current V_{CC}/r_c .

Figure 12-25 (a) Tuned class C amplifier; (b) voltage gain versus frequency; (c) dc equivalent circuit is unbiased; (d) two load lines; (e) ac equivalent circuit.

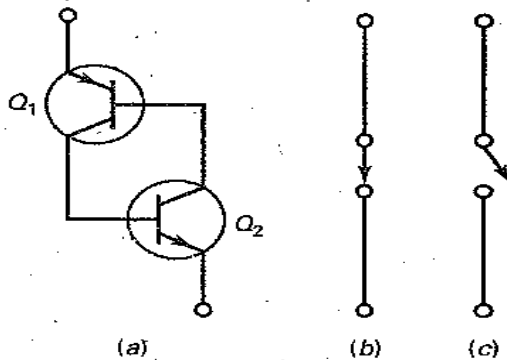


Thyristors

The four layer diode

Thyristor operation can be explained in terms of the equivalent circuit shown in Fig. 15-1a. The upper transistor Q_1 is a *pn*p device, and the lower transistor Q_2 is an *npn* device. The collector of Q_1 drives the base of Q_2 . Similarly, the collector of Q_2 drives the base of Q_1 .

Figure 15-1 Transistor latch.



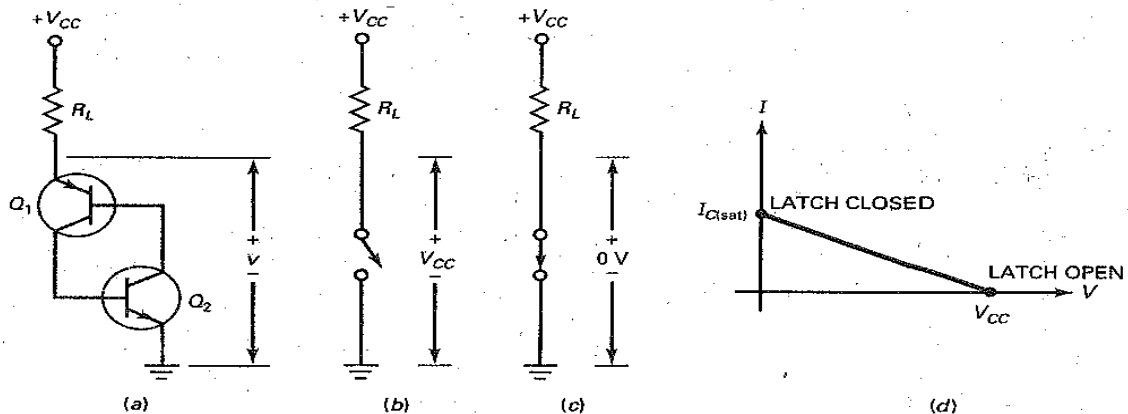
Positive feedback

The unusual connection of Fig. 15-1a uses *positive feedback*. Any change in the base current of Q_2 is amplified and fed back through Q_1 to magnify the original change. This positive feedback continues changing the base current of Q_2 until both transistors go into either saturation or cutoff.

For instance, if the base current of Q_2 increases, the collector current of Q_2 increases. This increases the base current of Q_1 and the collector current of Q_1 . More collector current in Q_1 will further increase the base current of Q_2 . This amplify-and-feedback action continues until both transistors are driven into saturation. In this case, the overall circuit acts like a closed switch (Fig. 15-1b).

On the other hand, if something causes the base current of Q_2 to decrease, the collector current of Q_2 decreases, the base current of Q_1 decreases, the collector current of Q_1 decreases, and the base current of Q_2 decreases further. This action continues until both transistors are driven into cutoff. Then, the circuit acts like an open switch (Fig. 15-1c).

Figure 15-2 Latching circuit.



Closing latch

The only way to close the latch of Fig. 15-2b is by **breakover**. This means using a large enough supply voltage V_{CC} to break down the Q_1 collector diode. Since the collector current of Q_1 increases the base current of Q_2 , the positive

feedback will start. This drives both transistors into saturation, as previously described. When saturated, both transistors ideally look like short circuits, and the latch is closed (Fig. 15-2c). Ideally, the latch has zero voltage across it when it is closed and the operating point is at the upper end of the load line (Fig. 15-2d).

How do we open the latch of Fig. 15-2a? By reducing the V_{CC} supply to zero. This forces the transistors to switch from saturation to cutoff. We call this type of opening **low-current drop-out** because it depends on reducing the latch current to a value low enough to bring the transistors out of saturation.

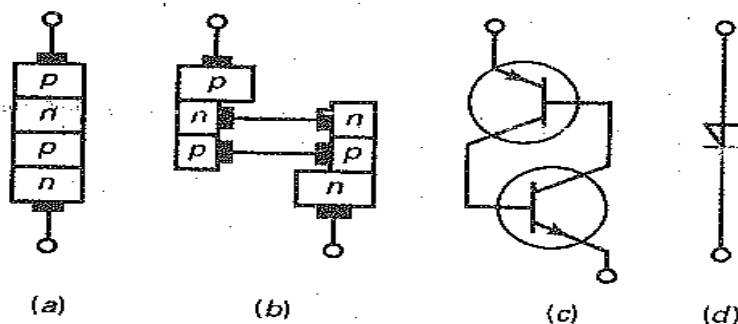
The Shockley diode

Figure 15-3a was originally called a **Schockley diode** after the inventor. Several other names are also used for this device: **four-layer diode**, **pnpn diode**, and **silicon unilateral switch (SUS)**. The device lets current flow in only one direction.

The easiest way to understand how it works is to visualize it separated into two halves, as shown in Fig. 15-3b. The left half is a **pnp transistor**, and the right half is an **nnp transistor**. Therefore, the four-layer diode is equivalent to the latch of Fig. 15-3c.

Figure 15-3d shows the schematic symbol of a four-layer diode. The only way to close a four-layer diode is by breakover. The only way to open it is by low-current drop-out, which means reducing the current to less than the holding current (given on data sheets). The holding current is the low value of current where the transistors switch from saturation to cutoff.

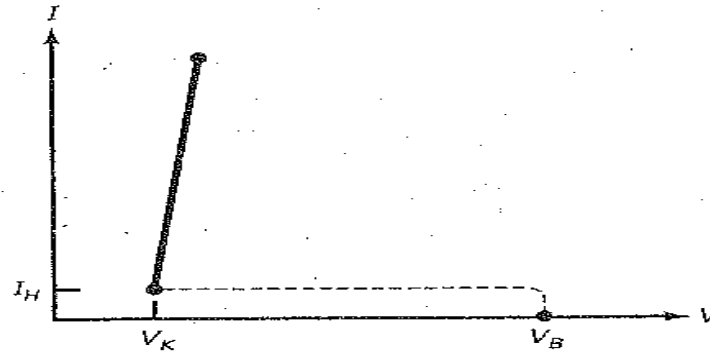
Figure 15-3 Four-layer diode.



Breakover characteristics

Figure 15-4 shows the graph of current versus voltage of a four-layer diode. The device has two operating regions: cutoff and saturation. The dashed line is the

Figure 15-4 Breakover characteristic.



transition path between cutoff and saturation. It is dashed to indicate that the device switches rapidly between the off and on states.

When the device is at cutoff, it has zero current. If the voltage across diode tries to exceed V_B , the device breaks over and moves rapidly along the dashed line to the saturation region. When the diode is in saturation, it is operating on the upper line. As long as the current through it is greater than the holding current I_H , the diode remains latched in the on state. If the current becomes less than I_H , the device switches into cutoff.

The silicon controlled rectifier(SCR)

The **SCR** is the most widely used thyristor. It can switch very large currents on and off. Because of this, it is used to control motors, ovens, air conditioners, and induction heaters.

By adding an input terminal to the base of Q_2 , as shown in Fig. 15-7a, we can create a second way to close the latch. Here is the theory of operation: When the latch is open, as shown in Fig. 15-7b, the operating point is at the lower end of the dc load line (Fig. 15-7d). To close the latch, we can couple a *trigger* (sharp pulse) into the base of Q_2 , as shown in Fig. 15-7a. The trigger momentarily increases the base current of Q_2 . This starts the positive feedback, which drives both transistors into saturation.

When saturated, both transistors ideally look like short circuits, and the latch is closed (Fig. 15-7c). Ideally, the latch has zero voltage across it when it is closed, and the operating point is at the upper end of the load line (Fig. 15-7d).

Figure 15-7 Transistor latch with trigger input.

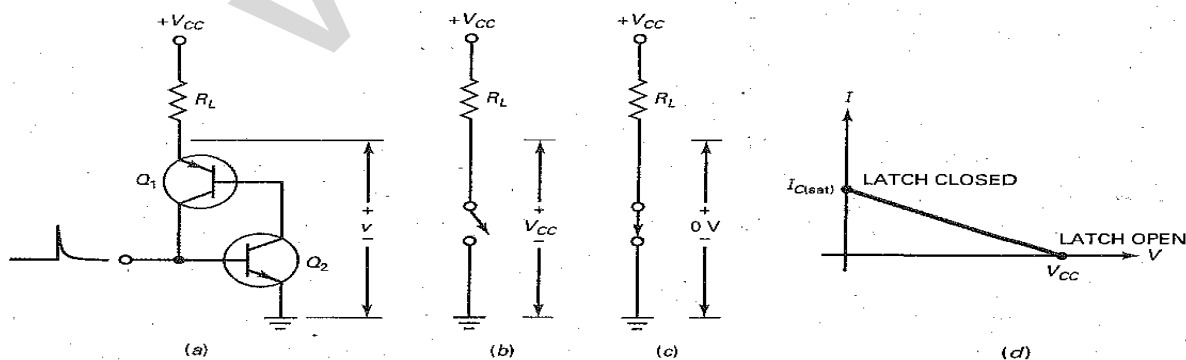
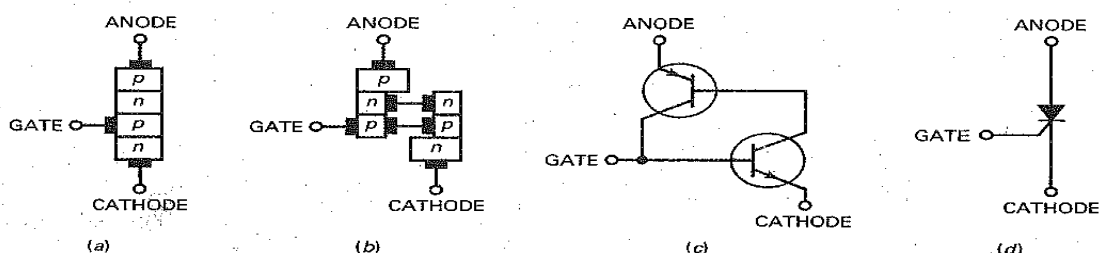


Figure 15-8 Silicon controlled rectifier (SCR).



Again, we can visualize the four doped regions separated into two transistors, as shown in Fig. 15-8*b*. Therefore, the SCR is equivalent to a latch with a trigger input (Fig. 15-8*c*). Schematic diagrams use the symbol of Fig. 15-8*d*. Whenever you see this symbol, remember that it is equivalent to a latch with a trigger input.

The SCR phase control

Figure 15-22 SCR phase control.

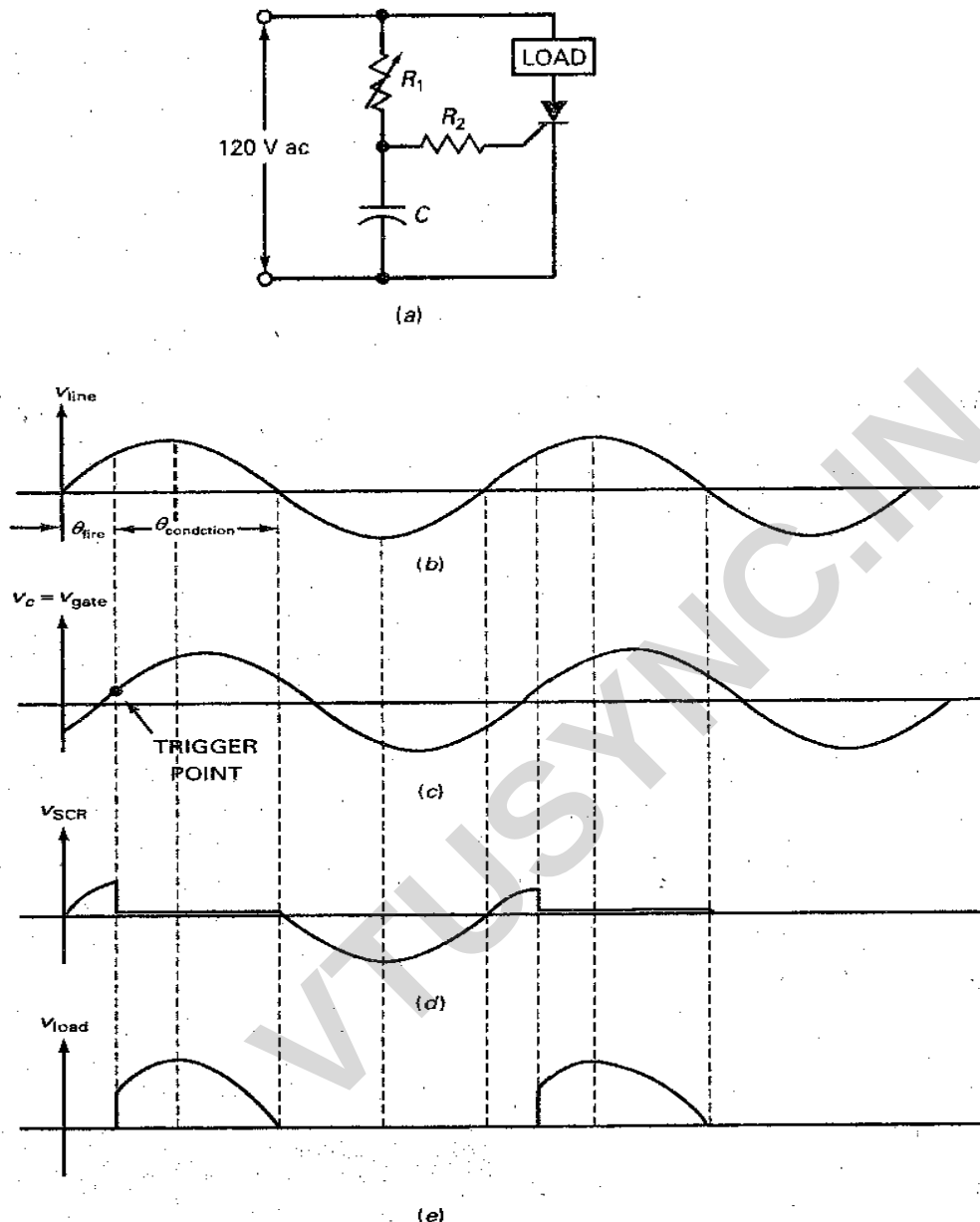


Figure 15-22*a* shows ac line voltage being applied to an SCR circuit that controls the current through a heavy load. In this circuit, variable resistor R_1 and capacitor C shift the phase angle of the gate signal. When R_1 is zero, the gate voltage is in phase with the line voltage, and the SCR acts like a half-wave rectifier. R_2 limits the gate current to a safe level.

When R_1 increases, however, the ac gate voltage lags the line voltage by an angle between 0 and 90°, as shown in Figs. 15-22*b* and *c*. Before the trigger point shown in Fig. 15-22*c*, the SCR is off and the load current is zero. At the trigger point, the capacitor voltage is large enough to trigger the SCR. When this happens, almost all of the line voltage appears across the load and the load current becomes high. Ideally, the SCR remains latched until the line voltage reverses polarity. This is shown in Figs. 15-22*c* and *d*.

The angle at which the SCR fires is called the **firing angle**, shown as θ_{fire} in Fig. 15-22a. The angle between the start and end of conduction is called the **conduction angle**, shown as $\theta_{\text{conduction}}$. The RC phase controller of Fig. 15-22a can change the firing angle between 0 and 90° , which means that the conduction angle changes from 180 to 90° .

To

determine the voltage across the capacitor, use the following steps:

First, find the capacitive reactance of C by:

$$X_C = \frac{1}{2\pi fc}$$

The impedance and phase angle of the RC phase-shift circuit is:

$$Z_T = \sqrt{R^2 + X_C^2} \quad (15-4)$$

$$\theta_Z = \angle - \arctan \frac{X_C}{R} \quad (15-5)$$

Using the input voltage as our reference point, the current through C is:

$$I_C \angle \theta = \frac{V_{\text{in}} \angle 0^\circ}{Z_T \angle - \arctan \frac{X_C}{R}}$$

Now, the voltage value and phase of the capacitor can be found by:

$$V_C = (I_C \angle \theta)(X_C \angle -90^\circ)$$

The amount of delayed phase shift will be the approximate firing angle of the circuit. The conduction angle is found by subtracting the firing angle from 180° .

Example 15-7

III MultiSim

Using Fig. 15-22a, find the approximate firing angle and conduction angle when $R = 26 \text{ k}\Omega$.

SOLUTION The approximate firing angle can be found by solving for the voltage value and its phase shift across the capacitor. This is found by:

$$X_C = \frac{1}{2\pi fc} = \frac{1}{(2\pi)(60 \text{ Hz})(0.1 \text{ }\mu\text{F})} = 26.5 \text{ k}\Omega$$

Because capacitive reactance is at an angle of -90° , $X_C = 26.5 \text{ k}\Omega \angle -90^\circ$.

Next, find the total RC impedance Z_T and its angle by:

$$Z_T = \sqrt{R^2 + X_C^2} = \sqrt{(26 \text{ k}\Omega)^2 + (26.5 \text{ k}\Omega)^2} = 37.1 \text{ k}\Omega$$

$$\theta_Z = \angle - \arctan \frac{X_C}{R} = \angle - \arctan \frac{26.5 \text{ k}\Omega}{26 \text{ k}\Omega} = -45.5^\circ$$

Therefore, $Z_T = 37.1 \text{ k}\Omega \angle -45.5^\circ$.

Using the ac input as our reference, the current through C is:

$$I_C = \frac{V_{in} \angle 0^\circ}{Z_T \angle \theta} = \frac{120 \text{ V}_{ac} \angle 0^\circ}{37.1 \text{ k}\Omega \angle -45.5^\circ} = 3.23 \text{ mA} \angle 45.5^\circ$$

Now, the voltage across C can be found by:

$$V_C = (I_C \angle \theta)(X_C \angle -90^\circ) = (3.23 \text{ mA} \angle 45.5^\circ)(26.5 \text{ k}\Omega \angle -90^\circ)$$

$$V_C = 85.7 \text{ V}_{ac} \angle -44.5^\circ$$

With the voltage phase shift across the capacitor of -44.5° , the firing angle of the circuit is approximately -45.5° . After the SCR fires, it will remain on until its current drops below I_H . This will occur approximately when the ac input is zero volts.

Therefore, the conduction angle is:

$$\text{conduction } \theta = 180^\circ - 44.5^\circ = 135.5^\circ$$

Bidirectional Thyristors

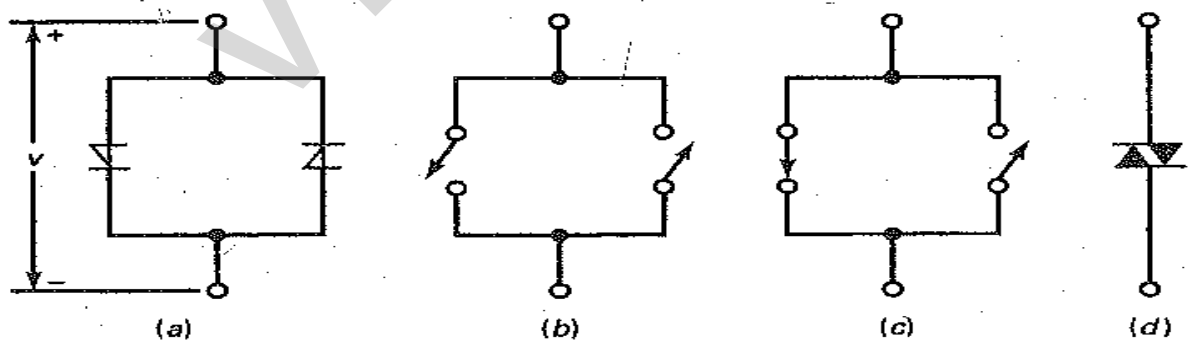
The two devices discussed so far, the four-layer diode and the SCR, are **unidirectional** because current can flow in only one direction. The **diac** and **triac** are **bidirectional thyristors**. These devices can conduct in either direction. The diac is sometimes called a *silicon bidirectional switch (SBS)*.

Diac

The diac can latch current in either direction. The equivalent circuit of a diac is two four-layer diodes in parallel, as shown in Fig. 15-24a, ideally the same as the latches in Fig. 15-24b. The diac is nonconducting until the voltage across it exceeds the breakover voltage in either direction.

For instance, if v has the polarity indicated in Fig. 15-24a, the left diode conducts when v exceeds the breakover voltage. In this case, the left latch closes, as shown in Fig. 15-24c. When v has the opposite polarity, the right latch closes. Figure 15-24d shows the schematic symbol for a diac.

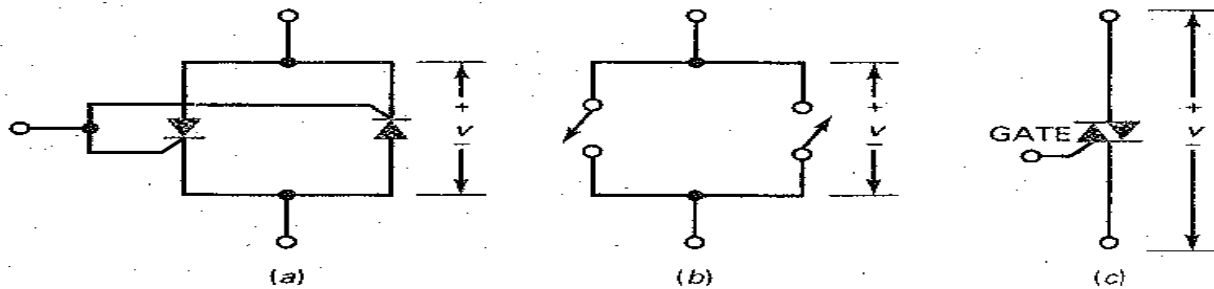
Figure 15-24 Diac.



Triac

The triac acts like two SCRs in reverse parallel (Fig. 15-25a), equivalent to the two latches of Fig. 15-25b. Because of this, the triac can control current in both directions. If v has the polarity shown in Fig. 15-25a, a positive trigger will close the left latch. When v has opposite polarity, a negative trigger will close the right latch. Figure 15-25c is the schematic symbol for a triac.

Figure 15-25 Triac.



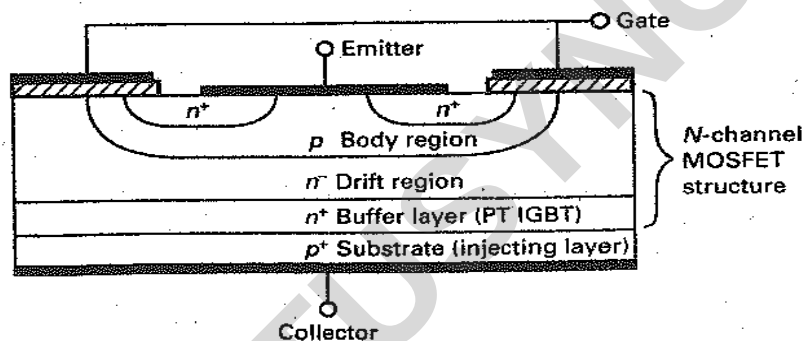
IGBTs(Insulated gate bipolar transistor)

Basic Construction

Power MOSFETs and BJTs can both be used in high-power switching applications. The MOSFET has the advantage of greater switching speed, and the BJT has lower conduction losses. By combining the low conduction loss of a BJT with the switching speed of a power MOSFET, we can begin to approach an ideal switch.

This hybrid device exists and is called an **insulated-gate bipolar transistor (IGBT)**. The IGBT has essentially evolved from power MOSFET technology. Its structure and operation closely resembles a power MOSFET. Fig. 15-30 shows the basic structure of an n -channel IGBT. Its structure resembles an n -channel power MOSFET constructed on a p -type substrate. As shown, it has gate, emitter, and collector leads.

Figure 15-30 Basic IGBT structure.



Two versions of this device are referred to as punch-through (PT) and nonpunch-through (NPT) IGBTs. Fig. 15-30 shows the structure of a PT IGBT. The PT IGBT has an $n+$ buffer layer between its $p+$ and $n-$ regions, and the NPT device has no $n+$ buffer layer.

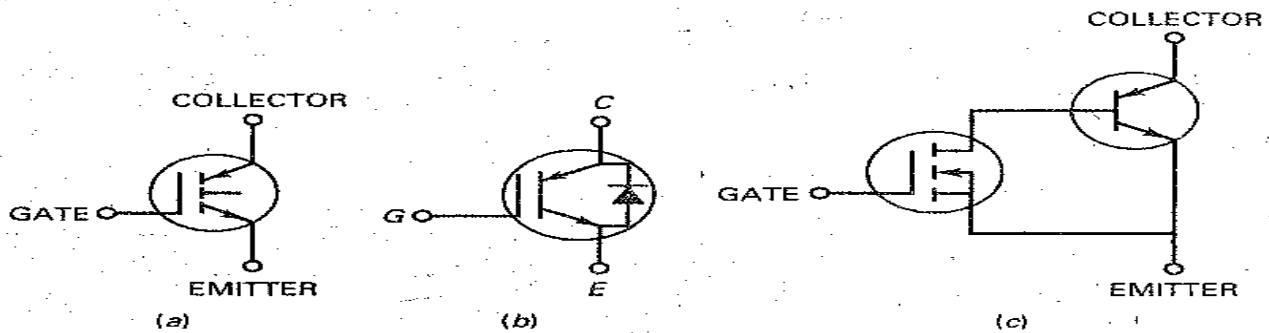
NPT versions have higher conduction $V_{CE(on)}$ values than PT versions and a positive temperature coefficient. The positive temperature coefficient makes the NPT suited for paralleling. PT versions, with the extra $n+$ layer, have the advantage of higher switching speeds. They also have a negative temperature coefficient.

IGBT Control

Figures 15-31a and b show two common schematic symbols for an n -channel IGBT. Also, Fig. 15-31c shows a simplified equivalent circuit for this device. As you can see, the IGBT is essentially a power MOSFET on the input side and a BJT on the output side. The input control is a voltage between the gate and emitter leads. The output is a current between the collector and emitter leads.

The IGBT is a normally off high-input impedance device. When the input voltage, V_{GE} , is large enough, collector current will begin to flow. This minimum voltage value is the gate threshold voltage, $V_{GE(th)}$.

Figure 15-31 IGBTs: (a) and (b) Schematic symbols; (c) simplified equivalent circuit.



Other Thyristors

Photo SCR

Figure 15-33 Photo-SCR.

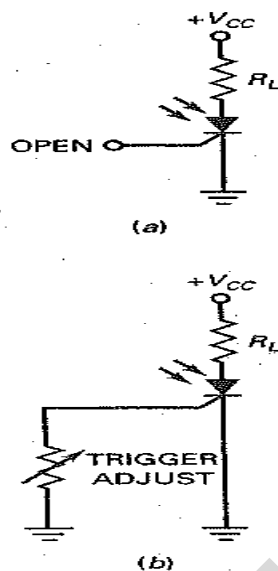


Figure 15-33a shows a *photo-SCR*, also known as a *light-activated SCR*. The arrows represent incoming light that passes through a window and hits the depletion layers. When the light is strong enough, valence electrons are dislodged from their orbits and become free electrons. The flow of free electrons starts the positive feedback, and the photo-SCR closes.

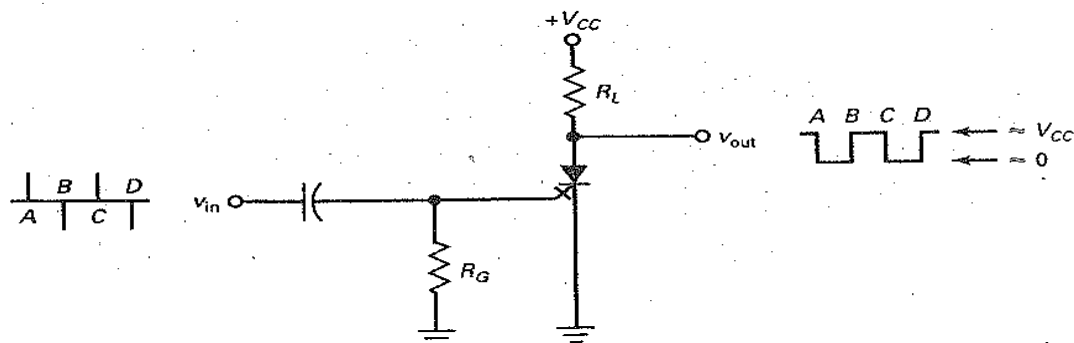
After a light trigger has closed the photo-SCR, it remains closed, even though the light disappears. For maximum sensitivity to light, the gate is left open, as shown in Fig. 15-33a. To get an adjustable trip point, we can include the trigger adjust shown in Fig. 15-33b. The resistance between the gate and ground diverts some of the light-produced electrons and reduces the sensitivity of the circuit to the incoming light.

Gate Controlled Switch

As mentioned earlier, low-current drop-out is the normal way to open an SCR. But the *gate-controlled switch* is designed for easy opening with a reverse-biased trigger. A gate-controlled switch is closed by a positive trigger and opened by a negative trigger.

Figure 15-34 shows a gate-controlled circuit. Each positive trigger closes the gate-controlled switch, and each negative trigger opens it. Because of this, we get the square-wave output shown. The gate-controlled switch has been used in counters, digital circuits, and other applications in which a negative trigger is available.

Figure 15-34 Gate-controlled switch.



Silicon Controlled Switch

Figure 15-35a shows the doped regions of a *silicon controlled switch*. Now an external lead is connected to each doped region. Visualize the device separated into two halves (Fig. 15-35b). Therefore, it's equivalent to a latch with access to both bases (Fig. 15-35c). A forward-bias trigger on either base will close the silicon controlled switch. Likewise, a reverse-bias trigger on either base will open the device.

Figure 15-35 Silicon controlled switch.

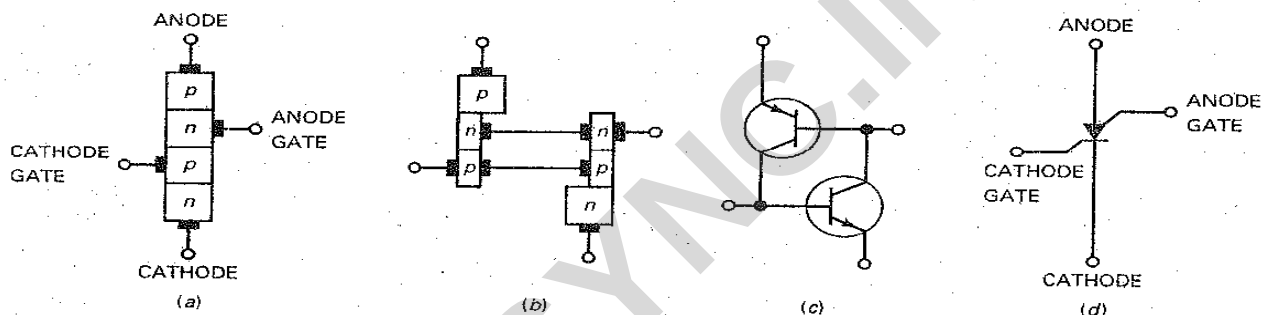


Figure 15-35d shows the schematic symbol for a silicon controlled switch. The lower gate is called the *cathode gate*, and the upper gate is the *anode gate*. The silicon controlled switch is a low-power device compared to the SCR. It handles currents in milliamperes rather than amperes.

Unijunction transistor

The **unijunction transistor (UJT)** has two doped regions, as shown in Fig. 15-36a. When the input voltage is zero, the device is nonconducting. When we increase the input voltage above the *standoff voltage* (given on a data sheet), the resistance between the *p* region and the lower *n* region becomes very small, as shown in Fig. 15-36b. Figure 15-36c is the schematic symbol for a UJT.

The UJT can be used to form a pulse-generating circuit called a **UJT relaxation oscillator**, as shown in Fig. 15-37. In this circuit, the capacitor charges toward V_{BB} . When the capacitor voltage reaches a value equal to the standoff voltage, the UJT turns on. The internal lower base (lower *n* region) resistance quickly drops in value allowing the capacitor to discharge. The capacitor discharge continues until low-current dropout occurs. When this happens, the UJT turns off and the capacitor begins to once again charge toward V_{BB} . The charging *RC* time constant is normally significantly larger than the discharge time constant.

The sharp pulse waveform developed across the external resistor at B_1 can be used as a trigger source for controlling the conduction angle of SCR and triac circuits. The waveform developed across the capacitor can be used in applications where a sawtooth generator is needed.

Figure 15-36 Unijunction transistor.

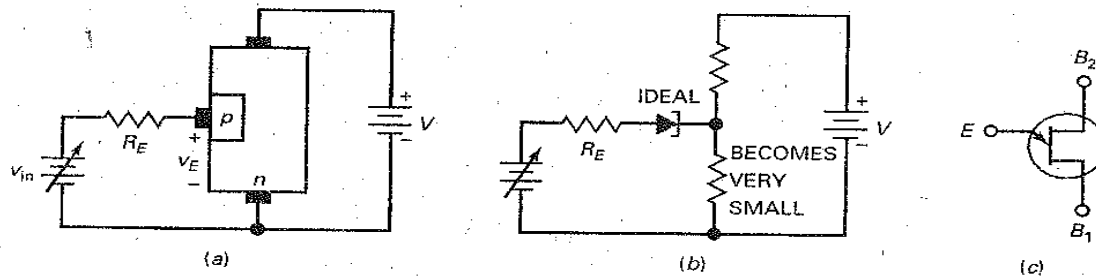
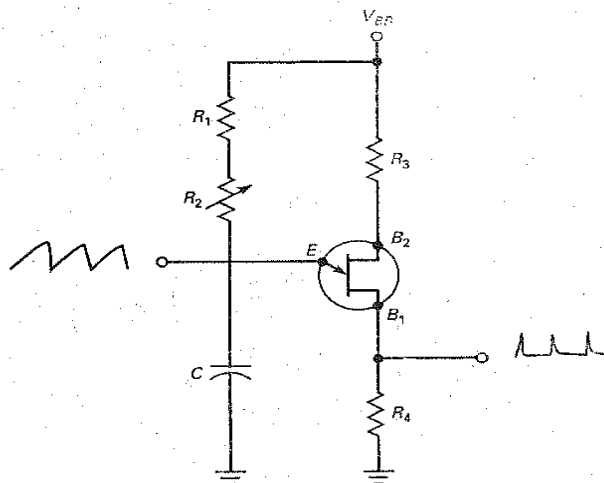


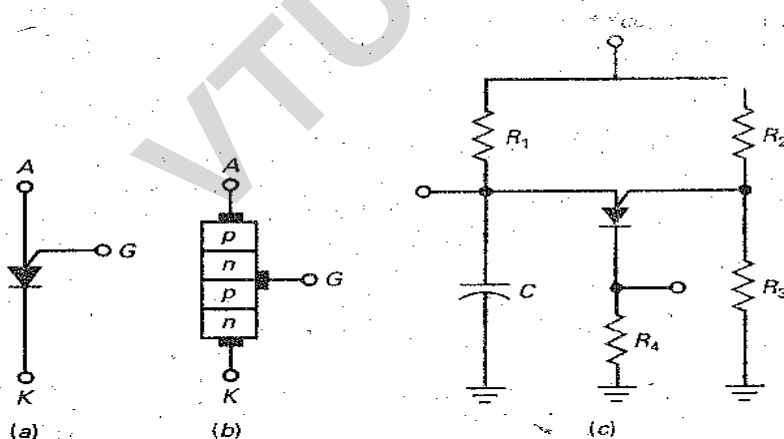
Figure 15-37 UJT relaxation oscillator.



Programmable unijunction transistor (PUT)

The **programmable unijunction transistor (PUT)** is a four-layer *pnpn* device, which is used to produce trigger pulses and waveforms similar to UJT circuits. The schematic symbol is shown in Fig. 15-38a.

Figure 15-38 PUT: (a) symbol; (b) structure; (c) PUT circuit.



Its basic construction, shown in Fig. 15-38b, is very different from a UJT, more closely resembling an SCR. The gate lead is connected to the *n* layer next to the anode. This *pn* junction is used to control the on and off states of the device. The cathode terminal is connected to a voltage point lower than the gate, typically at a ground point. When the anode voltage becomes approximately 0.7 V higher than the gate voltage, the PUT turns on. The device will remain in the on state until its anode current falls below the rated holding current, normally given as its valley current, I_V . When this happens, the device returns to its off state.

The PUT is considered to be programmable because the gate voltage can be determined by an external voltage divider. This is shown in Fig. 15-38c. The

external resistors, R_2 and R_3 , establish the gate voltage, V_G . By changing these resistor values, the voltage on the gate can be modified or “programmed,” thus changing the required anode voltage for firing. When the capacitor charges up through R_1 , it must reach a voltage value approximately 0.7 V higher than V_G . At that point, the PUT turns and the capacitor discharges. As with the UJT, sawtooth and trigger-pulse waveforms can be developed for controlling thyristors.

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